



SHRINIDHI

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PERSONAL PROFILE

Master's student in Computational Sciences in Engineering with hands-on experience across the CFD tool chain: mesh generation (Pointwise, AutoGrid5, Gmsh), turbulence modelling (Spalart-Allmaras, $k-\omega$ SST) in SU2-based high-lift studies, and Python automation of CFD workflows on Linux/HPC. Experienced in high-lift aerodynamics, unsteady simulations and development of in-house analysis tools within multidisciplinary research teams; familiar with conceptual aircraft design and optimization (UNICADO). Independent and target-oriented, with a track record of documenting and communicating complex technical results.

WORK EXPERIENCE

Master's Thesis | Turbomachinery Aeromechanics

June 2026 – Present

Everllence SE (formerly MAN Energy Solutions)

Augsburg, Germany

- Thesis: "Aerodynamic Excitation and Damping of Intentionally Mistuned Blisks in Axial Turbines"
- Numerically modeling aerodynamic excitation and damping of intentionally mistuned blisks in axial turbines; building an NLH-based operating-point database of complex harmonic pressures from idle to full load.
- Developing an Aerodynamic-Influence-Coefficient (AIC) based aerodynamic damping model for aeroelastic forced-response analyses.
- Developing scripts on a Linux workstation that automate the full workflow on HPC (meshing, speedline generation, job submission and monitoring, post-processing) for steady-state and rigid-/moving-mesh NLH runs, extended to mistuned blisks across multiple operating points.

Research Assistant (HiWi)

June 2024 – Present

Technische Universität Braunschweig - Institut für Flugzeugbau und Leichtbau

Germany

- Testing and documenting Python-based optimization loops within the UNICADO (UNiversity Conceptual Aircraft Design and Optimization) framework using Design of Experiments and Surrogate Modelling techniques.
- Extended WuFoil (Python tool integrating Gmsh, SU2, ML) with constrained optimization and support for unsteady aerodynamic flow analyses over different airfoil profiles.
- Supported lecture material development for "Design of Commercial Aircrafts I," covering aircraft sizing, aerodynamics, performance, and cost estimation in early-stage aircraft design

Research Assistant (HiWi)

Dec. 2024 – Jan. 2025

Technische Universität Braunschweig - Institut für Strömungsmechanik

Germany

- Set up droplet velocity and size measurement rig in icing wind tunnel for aircraft wing icing research, using Phase Doppler Anemometry and Interferometry (PDA/PDI) to characterize and validate multiphase flows.

Research Assistant (HiWi)

Aug. 2023 – Apr. 2024

Technische Universität Braunschweig - Institut für Flugzeugbau und Leichtbau

Germany

- Assisted in integrating Spalart-Allmaras and $k-\omega$ SST turbulence models into PyAeroSweep (Python-based aircraft aerodynamic analysis tool) for different high-lift configurations, including unsteady simulations.
- Supported the automation of mesh generation (structured/unstructured, Fidelity Pointwise) and SU2 CFD workflows on Linux; contributed to technical documentation

Intern

Aug. 2021 – Sept. 2021

Hindustan Aeronautics Limited - Aerospace Division

India

- Gained exposure to satellite launch vehicle assembly, welding, CNC machining, tool design, quality control, and cross-departmental aerospace manufacturing operations.

EDUCATION

Master of Science | Computational Sciences in Engineering

Oct. 2022 – Feb. 2027 (Expected)

Technische Universität Braunschweig

Braunschweig, Germany

Relevant Coursework: Fluid Mechanics, Finite-Volume-Method, High Lift Aerodynamics (incl. unsteady flow), Fluid-Structure Interaction, Flow-Induced Vibrations, Aerothermodynamics, Thermodynamics & Statistics, Multidisciplinary Design Optimization (MDO) & Applied MDO, ODE, PDE, Numerical Methods in ODEs & PDEs.

Summer School | Advanced Sustainable Aviation Technologies

Technische Universität Braunschweig

Aug. 2023

Braunschweig, Germany

*Relevant Coursework: Overall Aircraft Requirements, Aircraft Aerodynamics (incl. instationary phenomena & active flow control), Advanced Propulsion Systems.***Bachelor of Engineering | Mechanical Engineering**

Visvesvaraya Technological University

Aug. 2018 – June 2022

Mysuru, India

*Relevant Coursework: Fluid Mechanics, Heat Transfer, Thermodynamics, Kinematics of Machines, Dynamics of Machines, Computer Aided Modelling and Analysis.***PROJECTS****Aerodynamic Design of a Fowler Flap for a Hybrid Laminar Flow Control Airfoil** Jan. 2025 – Apr. 2025Studienarbeit - in association with Sustainable and Energy-Efficient Aviation (SE²A) Cluster

Germany

- Designed and optimized Fowler flap for HLFC airfoil using SU2 (k- ω SST) and automated Python-CFD workflow (PyAeroSweep), achieving up to **4.5% gain** in $C_{L_{max}}$.
- Conducted mesh and parameter sensitivity studies; analyzed unsteady flow detachment and reattachment in geometrically complex flap regions; applied systematic optimization for takeoff and landing configurations; post-processed results in ParaView.

Developing Methodology for Model Fitting in Fluid-Structure Coupled Problems

Jan. 2025

Course Project - in association with Chair of Structural Analysis, TU München

Germany

- Modeled Fluid-Structure Interaction of slender beam in wind tunnel using Kratos Multiphysics; developed optimization workflow for material parameter identification, evaluating coupling strategies (Gauss-Seidel, Jacobi) in unsteady flow contexts.

Design and Fabrication of Quad-Plane for All-Terrain Applications

Oct. 2021 – June 2022

Bachelor Thesis

India

- Led four-person team to design, fabricate and flight-test fixed-wing eVTOL UAV (1.2 m wingspan, 3 kg MTOW) optimized for medical supply delivery.
- Performed initial aerodynamic design in XFLR5, CAD modeling in SolidWorks, CFD-based optimization and structural validation in Ansys Workbench; designed and 3D-printed propeller ducts.

SAE International - SAE Aero Design West 2022 - Vymanika

June 2021 – Apr. 2022

Team Captain - SAE Aero Design Competition

CA, USA

- Captain of 12-member team, leading all project phases for RC aircraft (1.2 m wingspan, 3 kg MTOW): design, aerodynamic and structural analysis, simulation, fabrication, and competition.
- Managed full-scale build and integration; performed CAD in SolidWorks, aerodynamic modeling in OpenVSP/XFLR5, CFD and FEA in Ansys Workbench; co-authored Technical Design Report.

SAEINDIA - Aerothon 2021 - Virtual Aero Design Contest - Vymanika

Apr. 2021 – June 2021

Team Captain - SAE Aero Design Competition

India

- Captain of seven-member team leading all aspects of UAV design, aerodynamic and structural analysis, and simulation (5.28 m wingspan, 25 kg MTOW).
- Lead CAD designer and aerodynamicist; developed designs in SolidWorks/XFLR5 and performed CFD and structural optimization in Ansys Workbench.

SKILLS**Languages:** English (Fluent/C1), German (DSH-1/B2), Hindi (Fluent), Kannada (Native).**Programming & Automation:** Python (CFD workflow automation, post-processing, optimization), MATLAB & Simulink (Basic), C/C++ (Basic).**CFD & Simulation:** NUMECA Fine/Turbo, SU2, Ansys Fluent, XFLR5, Kratos Multiphysics.**Mesh Generation:** AutoGrid5, Fidelity Pointwise, Gmsh, Ansys Meshing.**CAD & Geometry:** CATIA V5 (Aerospace Design), SolidWorks (Certified Professional), 3DEXPERIENCE (Basic), OpenVSP.**Post-processing:** CFView, ParaView, Ansys CFD-Post, Python visualization libraries.**Tools & Platforms:** LaTeX, Git, MS Office Suite, Google Workspace, Windows, Linux, HPC.**VOLUNTEERING****Rotaract Club of Vidyavardhaka College of Engineering**

July 2021 – June 2022

Treasurer

India

- Coordinated events and awareness campaigns on entrepreneurship, mental health, and community initiatives.

NSS and Red Cross Youth Wing

Dec. 2018 – July 2022

Volunteer

India

- Coordinated outreach in villages, including surveys, awareness campaigns, and empowerment initiatives.